

Math 141, F11
Exam 2

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	24	
2	24	
3	26	
4	26	
Total	100	

Good Luck !

Problem 1. (24 points) Find each requested derivative. Be sure to simplify your results.

(a) Find $f'(x)$ when $f(x) = \frac{1}{x^2} + \sqrt[3]{x^3}$

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left[x^{-2} + x^{\frac{3}{2}} \right] \\ &= -2x^{-3} + \frac{3}{2}x^{\frac{1}{2}} \\ &= -\frac{2}{x^3} + \frac{3}{2}\sqrt{x} \end{aligned}$$

(b) Find $f'(x)$ when $f(x) = \sin x \cos x$

$$\begin{aligned} f'(x) &= \sin x (-\cos x) + \cos x \cdot \cos x \\ &= \cos^2 x - \sin^2 x \\ &= \cos 2x \end{aligned}$$

(c) Find $\frac{dy}{dx}$ when $y = \frac{x}{x^2 + 9}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1 \cdot (x^2 + 9) - x(2x)}{(x^2 + 9)^2} \\ &= \frac{-x^2 + 9}{(x^2 + 9)^2} \end{aligned}$$

(d) Find $\frac{dy}{dx}$ when $y = 3^x - e^x$

$$\frac{dy}{dx} = 3^x \ln 3 - e^x$$

Problem 2. (24 points) Find the derivative of each of the following functions.

(a) $f(x) = \left(\frac{x^2 + 1}{x^2 - 1} \right)^{10}$

$$\begin{aligned} f'(x) &= 10 \left(\frac{x^2 + 1}{x^2 - 1} \right)^9 \frac{d}{dx} \left[\frac{x^2 + 1}{x^2 - 1} \right] \\ &= 10 \left(\frac{x^2 + 1}{x^2 - 1} \right)^9 \frac{2x(x^2 - 1) - (x^2 + 1) \cdot 2x}{(x^2 - 1)^2} \\ &= 10 \left(\frac{x^2 + 1}{x^2 - 1} \right)^9 \frac{-4x}{(x^2 - 1)^2} \\ &= -40 \frac{x(x^2 + 1)^9}{(x^2 - 1)^{11}} \end{aligned}$$

(b) $f(x) = xe^{-x^3}$

$$\begin{aligned} f'(x) &= x \cdot e^{-x^3} \cdot (-3x^2) + e^{-x^3} \\ &= e^{-x^3} (1 - 3x^3) \end{aligned}$$

(d) $f(x) = \log_2(x\sqrt{x-1})$

$$\begin{aligned} f(x) &= \log_2 x + \log_2 \sqrt{x-1} = \log_2 x + \frac{1}{2} \log_2 (x-1) \\ \Rightarrow f'(x) &= \frac{1}{x \ln 2} + \frac{1}{2} \cdot \frac{1}{(x-1) \ln 2} \cdot 1 \\ &= \frac{1}{x \ln 2} + \frac{1}{2(x-1) \ln 2} \end{aligned}$$

(c) $f(x) = \tan^{-1}(x^2)$

$$\begin{aligned} f'(x) &= \frac{1}{1 + (x^2)^2} \cdot 2x \\ &= \frac{2x}{1 + x^4} \end{aligned}$$

Problem 3. (26 points)

- (a) Using the **logarithmic differentiation** to find the $\frac{dy}{dx}$ if $y = (x^2 + 3)^{\sin x}$.

Solution: $\ln y = \ln (x^2 + 3)^{\sin x}$

$$\ln y = \sin x \ln (x^2 + 3)$$

$$\frac{d}{dx} [\ln y] = \frac{d}{dx} [\sin x \ln (x^2 + 3)]$$

$$\frac{1}{y} \frac{dy}{dx} = \sin x \cdot \frac{2x}{x^2 + 3} + \cos x \ln (x^2 + 3)$$

$$\frac{dy}{dx} = y \left[\frac{2x \sin x}{x^2 + 3} + \cos x \ln (x^2 + 3) \right]$$

$$= (x^2 + 3)^{\sin x} \left[\frac{2x \sin x}{x^2 + 3} + \cos x \ln (x^2 + 3) \right]$$

- (b) A super-ellipse is defined implicitly by the equation $16x^4 + y^4 = 32$. Use the **implicit differentiation** to find $\frac{dy}{dx}$, and then based this result find an equation for the tangent line to the super-ellipse at the point $(1, 2)$.

Solution: $\frac{d}{dx} [16x^4 + y^4] = \frac{d}{dx} [32]$

$$64x^3 + 4y^3 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{64x^3}{4y^3}$$

$$= -\frac{16x^3}{y^3}$$

At the point $(1, 2)$, we have the derivative

$$\left. \frac{dy}{dx} \right|_{x=1, y=2} = -\frac{16 \cdot 1^3}{2^3} = -2$$

Thus the equation for the tangent line is given by

$$\frac{y-2}{x-1} = -2$$

$$y-2 = -2x+2 \Rightarrow 2x+y-4=0$$

Problem 4. (26 points) Let $f(x) = x^4 - 8x^2 + 4$.

(a) Find all critical numbers of f , and then determine the intervals on which f is increasing or decreasing.

Solution: $f'(x) = 4x^3 - 16x = 4x(x^2 - 4) = 4x(x-2)(x+2)$

critical numbers: $f'(x) = 0$

$\Rightarrow x = 0, x = 2, x = -2$.

Intervals	$f'(x)$	$f(x)$
$(-\infty, -2]$	-	decreasing
$[-2, 0]$	+	increasing
$[0, 2]$	-	decreasing
$[2, +\infty)$	+	increasing

(b) Find the absolute maximum and absolute minimum values of f on the interval $[-3, 3]$.

Solution: At the critical numbers, we have

$f(0) = 4, f(-2) = -12, f(2) = -12$

At the two end points $-3, 3$, we have

$f(-3) = 13, f(3) = 13$

Absolute maximum value: 13

Absolute minimum value: -12.

(c) Find the inflection points of f , and then determine intervals on which f is concave upward or concave downward.

Solution: $f''(x) = 12x^2 - 16 = 4(3x^2 - 4) = 4(\sqrt{3}x - 2)(\sqrt{3}x + 2)$

$f''(x) = 0 \Rightarrow x = \frac{2}{\sqrt{3}} \text{ or } x = -\frac{2}{\sqrt{3}}$

Intervals	$f''(x)$	$f(x)$
$(-\infty, -\frac{2}{\sqrt{3}}]$	+	Concave upward
$[-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}]$	-	Concave downward.
$[\frac{2}{\sqrt{3}}, +\infty)$	+	Concave upward.

Inflection points:

$(-\frac{2}{\sqrt{3}}, -\frac{44}{9})$

$(\frac{2}{\sqrt{3}}, -\frac{44}{9})$

Extra Credits. (5 points) Find

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^2(2x)} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{\sin^2(2x)} \\ &= \left(\lim_{x \rightarrow 0} \frac{x}{\sin 2x} \right) \left(\lim_{x \rightarrow 0} \frac{x}{\sin 2x} \right) \\ &= \left(\frac{1}{2} \lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \right) \left(\frac{1}{2} \lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \right) \\ &= \frac{1}{2} \cdot 1 \cdot \frac{1}{2} \cdot 1 \\ &= \frac{1}{4} \end{aligned}$$